QA Automation Engineer (Junior)

Homework Assignment

Time: ~4 hours. Please do not spend more — we grade against a 4-hour scope, and “unfinished but honest” scores better than “polished but 12 hours.”

Deadline: 7 days from receipt

Target system: https://stage.longo.lv — our staging environment. No login required for anything in this task.

Access: stage.longo.lv is reachable **only from Latvian IP addresses.** If you are currently outside Latvia, or your ISP/mobile operator routes you through a non-LV address, tell us before you start and we will arrange access. Do not lose an hour of your four to a connection problem — just email us.

Follow-up: a 30-minute call where you run your tests live on your machine and walk us through the code.

Ground rules (please read — these matter to us)

You are working against staging, not production. That means you can be more adventurous than you could on a live site, but a few things still apply:

- **Never point your tests at longo.lv (production).** Keep the base URL in configuration so there is no chance of an accidental run against the live site — we will look for this.

2. **Do not request SMS / OTP login codes.** Staging is not wired to a real SMS gateway. Form submissions other than login are fine; please prefix any free-text field with **QA TEST** so we can identify and clean up your data.
3. **Be reasonable with traffic.** No load or stress testing. Maximum 2 parallel workers. Do not crawl more than ~2–3 pages of catalog results.
4. **Staging is staging.** Data, stock and prices differ from production, the environment may be redeployed while you work, and some features may be mid-development. If something looks broken, report it — deciding whether an anomaly is a real defect or an artefact of the environment is part of the job. Say which one you think it is and why.
5. **AI tools are allowed and expected.** Claude, Copilot, ChatGPT, Cursor — use whatever you normally use. We care that you can explain and defend every line you submit, not that you typed it yourself.
6. Your code stays yours. We will not use it in our products.

Part 1 — Automation (~2.5 hours)

Pick any stack you are comfortable with — for example Playwright, Selenium, or Cypress with Python, JavaScript/TypeScript, Java, or C#. **Tell us in the README why you picked it.** There are no bonus points for picking something exotic.

Required UI scenarios

1. **Catalog filtering.** Open the car catalog, apply a filter combination (e.g. make + body type + a price range). Assert that the results *actually match the filter* — check at least two attributes on the first few result cards. "Some results appeared" is not a passing assertion.

Negative / empty state. Choose a filter combination that returns zero or very few results. Assert the empty-state behaviour is correct.

Card → detail page consistency. Open the first result from the catalog and assert that key data on the detail page matches what the card showed (price, model, year, mileage).

Language switching. Switch between locales (lv / en / ru). Assert the locale change is reflected in the URL and in visible content, and that it persists across a page navigation.

Required non-UI scenario

One check below the UI level. Your choice — for example: intercept the network request the catalog makes and validate its response (status, structure, that the filter parameter was actually applied); or verify that a set of car detail URLs return 200 with a title and meta description; or compare the data rendered on a detail page against the data in the API response behind it. We want to see that you don't think testing stops at the browser.

Engineering expectations

7. Page Object Model or equivalent separation — no selectors scattered through test bodies
8. Proper waits, not fixed `sleep()` calls, wherever a proper wait is possible
9. Base URL, browser, and locale configurable rather than hardcoded
10. Tests independent of each other and runnable with a **single documented command**
11. A `README.md` covering: stack choice and why, how to install, how to run, what's covered, what you deliberately left out, and known flakiness

Optional extras (genuinely optional — only if you have time left): HTML report, parallel execution, GitHub Actions workflow, Dockerfile, visual or accessibility checks.

Part 2 — Manual testing (~1 hour)

Exploratory session (40 min, timeboxed). Charter: *"the financing / monthly payment calculator and car filtering, on a mobile viewport."* Deliver a short session note: what you covered, what you consciously did not cover, and what you found.

Bug reports. Report every defect or inconsistency you found (in Part 1 or Part 2) in a table with: ID, title, preconditions, steps to reproduce, expected result, actual result, severity **and** priority with a one-line justification for each, environment (browser / OS / viewport), and evidence (screenshot or short recording). Because staging data changes, always include the **exact URL** and the **date and time** you observed the issue. *If you found no bugs, say so explicitly and explain where you looked.*

Test cases. Write 5 test cases for one feature of your choice, in any format you like. Include at least 2 negative cases and 1 boundary case.

Part 3 — AI usage and reflection (~30 minutes)

`AI_USAGE.md` — which tools you used, 2–3 of the prompts that worked best for you, and **one concrete example where the AI gave you something wrong, fragile, or subtly incorrect, plus how you noticed and fixed it.** This is one of the most important parts of the assignment for us.

Half a page: if you had three more days on this system, what would you automate next, and what would you deliberately keep as manual testing? Why?

What to submit

A link to a public Git repository (or a zip) containing:

12. The automation project, runnable per the README
13. `README.md`
14. Bug report + exploratory session notes + test cases (Markdown, PDF, or a spreadsheet — your choice)
15. `AI_USAGE.md` + your reflection

We are hiring at junior level. We do not expect a perfect framework. We expect clear thinking, honest scope, and someone we can have a good technical conversation with on the follow-up call.

Questions during the assignment are welcome — email viktorija.naujoke@longo.group. Asking a good clarifying question is a positive signal, not a negative one.